

Recombining trees for one-dimensional forward rate models

Dariusz Gatarek,
Deloitte&Touche
and Polish Academy of Sciences
dgatarek@deloitteCE.com
and
Jarosław Kołakowski
SAS Institute

There exist two classes of interest rate models. Short rate models (HW, CIR, BDT), easy in pricing and tough in calibration and forward rate models (HJM, BGM), easy in calibration and tough in pricing. Parameters in short rate models have no natural interpretation in terms of market volatility but many options can be priced on recombining trees. We find particularly inconvenient the procedure of fitting the initial yield curve – necessary for many short rate models. Parameters of forwards rate models (especially BGM) have direct link to market volatility but there exists a common prejudice that recombining trees cannot be applied to forward rate models. This paper is an attempt to construct a model allowing both recombining trees and “calibration without programming”. We would like to call both presented models “simplest possible term structure models” – at least we do not know any simpler model. We strongly believe in a need for both: complex and universal models and simple ones. Forward rate models can be suited to be either complex or simple because of their rich and flexible volatility structure – “...it eats everything you feed it on...” [BM, p.495]. Tree-pricing of exotic options is particularly useful in two cases:

1. VaR calculations.

All options in all models can be priced by Monte Carlo but there exist situations where Monte Carlo is not practical. The concept of Monte Carlo (for derivatives pricing) within Monte Carlo (for VaR calculations) cannot be taken seriously. Therefore a faster (maybe less accurate) model is needed.

2. Bermudan swaptions.

As it was shown in [AA, BW] prices of Bermudan swaptions – probably the most important exotic products – are not sensitive to the number of model factors. Here tree-pricing is not only fast but also accurate.

On the other hand there exist derivatives sensitive to the number of factors – spread options for instance – and single factor models should not be applied there.

1. HJM model with the flavour of BGM

Let δ be accrual period for both interest rates and swaps. We assume it constant for simplicity. Define consecutive swap points as $T_{i+1} = T_i + \delta$ for certain initial $T = T_0 < \delta$. Forward discount factors are defined as

$$(1.1) \quad D_n(t) = \frac{B(t, T_{n-1})}{B(t, T_n)},$$

forward LIBOR rates as $L_n(t) = \delta^{-1}(D_n(t) - 1)$ and forward swap rates as

$$(1.2) \quad S_{nN}(t) = \frac{\sum_{i=n+1}^N B(t, T_i) L_i(t)}{\sum_{i=n+1}^N B(t, T_i)} = \frac{B(t, T_n) - B(t, T_N)}{\delta \sum_{i=n+1}^N B(t, T_i)},$$

where $B(t, T)$ are discount factors on the period $[t, T]$. Assume that $B(t, T)$ are differentiable and define forward instantaneous rates by

$$f(t, x) = -\frac{\partial}{\partial x} \ln B(t, x).$$

In the Heath-Jarrow-Morton (HJM) model the forward instantaneous rates satisfy

$$df(t, x) = \sigma(t, x) \int_t^x \sigma(t, u) du dt + \sigma(t, x) dW(t),$$

where $\sigma(t, x)$ is a deterministic volatility process and $W(t)$ a Wiener process. Discount factors satisfy the equation

$$dB(t, x) = -f(t, t)B(t, x)dt - B(t, x) \int_t^x \sigma(t, u) du dW(t)$$

Define

$$\gamma(t, x) = \int_{x-\delta}^x \sigma(t, u) du$$

and assume that $\gamma(t, x) = \gamma(t)$. In the single factor HJM model forward discount factors $D_n(t)$ follow the equation

$$(1.3) \quad dD_n(t) = \left[\delta^{-1} (T_n - t) \right]^+ \gamma^2(t) D_n(t) dt + \gamma(t) D_n(t) dW(t).$$

There are two ways of proving (1.3): either to use directly Ito formula or to consider the model as displaced Libor market model with displacement equal δ^{-1} [R]. All calculations are left to the reader.

2. Calibration to caps/floors or co-terminal swaptions

The model can be easily calibrated to caps/floors or co-terminal swaptions. Let φ_n be the volatility of an at-the-money (ATM) caplet on the period $[T_n, T_{n+1}]$. ATM caplet price is given by:

$$(2.1) \quad Caplet_{n+1}(0) = \delta B(0, T_{n+1}) \left(L_{n+1}(0) N\left(\frac{\sqrt{\theta_n}}{2}\right) - L_{n+1}(0) N\left(-\frac{\sqrt{\theta_n}}{2}\right) \right),$$

where

$$(2.2) \quad \theta_n = T_n \varphi_n^2,$$

$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-u^2/2} du.$$

Calibration to caplet prices consists of finding sequence of parameters σ_n satisfying

$$(2.3) \quad \text{Caplet}_{n+1}(0) = B(0, T_{n+1}) \left(D_{n+1}(0) N\left(\frac{\sqrt{\sigma_n}}{2}\right) - D_{n+1}(0) N\left(-\frac{\sqrt{\sigma_n}}{2}\right) \right).$$

Since $\delta L_n(t) + 1 = D_n(t)$, (2.3) reduces to

$$\begin{aligned} k_n &= D_{n+1}(0) \left(N\left(\frac{\sqrt{\sigma_n}}{2}\right) - \frac{1}{2} \right) = (1 + \delta L_{n+1}(0)) \left(N\left(\frac{\sqrt{\sigma_n}}{2}\right) - \frac{1}{2} \right) \\ &= \delta L_{n+1}(0) \left(N\left(\frac{\sqrt{\theta_n}}{2}\right) - \frac{1}{2} \right) \end{aligned}$$

Since $\sigma_{n-1} \leq \sigma_n \leq \theta_n$ and $N'(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ the equation (2.3) can be easily solved by Newton method or bisection. Since σ_n are very small, the one-step Newton method

$$\sqrt{\sigma_n} = \sqrt{\sigma_{n-1}} + 2 \left(\frac{k_n}{D_{n+1}(0)} - \frac{k_{n-1}}{D_n(0)} \right) e^{\sigma_{n-1}/8} \sqrt{2\pi}$$

is good. Taking two or three steps makes the approximation almost perfect. By (1.3)

$$\begin{aligned} D_n(T_k) &= D_n(0) \exp \left\{ \int_0^{T_k} \gamma(s) dW(s) + \int_0^{T_k} \gamma^2(s) \left([\delta^{-1}(T_n - s)] - \frac{1}{2} \right) ds \right\} \\ &= D_n(0) \exp \left\{ \int_0^{T_k} \gamma(s) dW(s) + \sum_{j=1}^n \min\{\sigma_k, \sigma_{j-1}\} - \frac{\sigma_k}{2} \right\} \end{aligned}$$

For the swap rate:

$$\delta S_{nN}(t) + 1 = \frac{\sum_{i=n+1}^N B(t, T_i) D_i(t)}{\sum_{i=n+1}^N B(t, T_i)} = \exp \left\{ \int_0^t \gamma(s) dW(s) - \frac{1}{2} \int_0^t \gamma^2(s) ds \right\} \bar{A}(t),$$

where

$$A_i(t) = D_i(0) \exp \left\{ \int_0^t \gamma^2(s) [\delta^{-1}(T_i - s)] ds \right\}$$

and

$$\bar{A}(t) = \frac{\sum_{i=n+1}^N B(t, T_i) A_i(t)}{\sum_{i=n+1}^N B(t, T_i)}$$

By the Ito formula:

$$d(\delta S_{nN}(t) + 1) = (\delta S_{nN}(t) + 1)((1 - F_{nN}(t))\gamma(t)dW(t) + drift \cdot dt),$$

where

$$F_{nN}(t) = \frac{\sum_{j=n+1}^N \sum_{i=n+1}^N B(t, T_j) B(t, T_i) (A_i(t) - A_j(t)) (i - n)}{\left(\sum_{i=n+1}^N B(t, T_i) \right)^2 \bar{A}(t)} = \frac{\sum_{i=n+1}^N B(t, T_i) (A_i(t) - \bar{A}(t)) (i - n)}{\sum_{i=n+1}^N B(t, T_i) \bar{A}(t)}.$$

If we assume that $B(t, T) \leq B(t, S)$ for $T \geq S$ (it holds in practice)

$$|F_{nN}(t)| \leq \max_{i,j} (A_i(t) - A_j(t)) \frac{N - n + 1}{2}$$

For flat initial yield curve $F_{nN}(t)$ is close to 0. Therefore ATM swaptions may be priced with the Black formula:

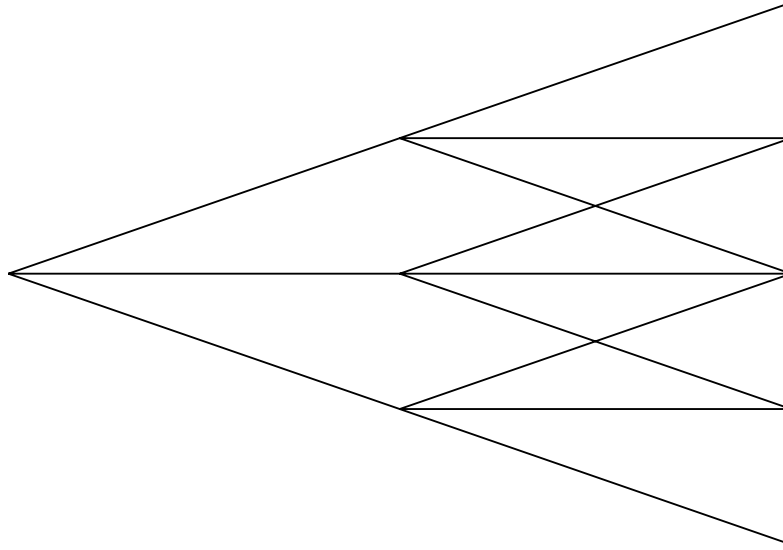
$$Swaption_{nN}(0) \cong \sum_{i=n+1}^N B(0, T_i) \left((\delta S_{nN}(0) + 1) N \left(\frac{\sqrt{\sigma_{nN}}}{2} \right) - (\delta S_{nN}(0) + 1) N \left(\frac{-\sqrt{\sigma_{nN}}}{2} \right) \right),$$

where $\sigma_{nN} = (1 - F_{nN}(0))^2 \sigma_n$ and

$$F_{nN}(0) = \frac{\sum_{i=n+1}^N B(0, T_i) (A_i(0) - \bar{A}(0)) (i - n)}{\bar{A}(0) \sum_{i=n+1}^N B(0, T_i)} = \frac{\delta \sum_{i=n+1}^N B(0, T_i) (L_i(0) - S_{nN}(0)) (i - n)}{(1 + \delta S_{nN}(0)) \sum_{i=n+1}^N B(0, T_i)}.$$

In conclusion: the model is easy to calibrate either to caps/floors or to co-terminal swaptions.

3. Trinomial trees



Random variables $\int_{T_{k-1}}^{T_k} \gamma(s) dW(s)$ can be approximated by discrete random variables Z_k . Let

$$p_0 = \frac{1}{2a^2} \int_0^{T_0} \gamma^2(s) ds = \frac{\sigma_0}{2a^2},$$

$$p_k = \frac{1}{2a^2} \int_{T_{k-1}}^{T_k} \gamma^2(s) ds = \frac{\sigma_k - \sigma_{k-1}}{2a^2},$$

where $a = \max_k \sqrt{\sigma_k - \sigma_{k-1}}$ and let $Z_{k,k=0,\dots,N}$ be independent random variables on the set $\{-a, 0, a\}$ defined by probabilities

$$P(Z_k = \pm a) = p_k,$$

$$P(Z_k = 0) = 1 - 2p_k.$$

Then the process $\sum_{i=0}^k Z_i$ is a Markov process on the net $\{\dots, -2a, -a, 0, a, 2a, \dots\}$ such that

$$E\left\{\sum_{i=0}^k Z_i\right\} = E\left\{\int_0^{T_k} \gamma(s) dW(s)\right\} = 0,$$

$$E\left\{\sum_{i=0}^k Z_i\right\}^2 = E\left\{\int_0^{T_k} \gamma(s) dW(s)\right\}^2.$$

4. Pricing of Bermudan options

Let $F(T_k, D(T_k))$ be an intrinsic value of an option, where $D(T_k) = \{D_1(T_k), \dots, D_N(T_k)\}$. We say that the option with intrinsic value $F(k, D(T_k))$ is Bermudan or mid-Atlantic if it can exercised only at Markov times from the set $\{T_n, T_{n+1}, \dots, T_{N_0}\}$, where $0 \leq n \leq N_0 \leq N$. By the general theory its value $V(0, L)$ at time 0 is equal

$$V(0, L) = \sup_{n \leq \tau \leq N_0} E\left\{F(T_\tau, D(T_\tau))B(0, T_0) \prod_{i=1}^{\tau} D_i(T_{i-1})^{-1} \mid D(0) = D\right\}.$$

Approximating $\int_{T_{k-1}}^{T_k} \gamma(s) dW(s)$ by Z_k we get

$$(4.1) \quad \begin{aligned} D_n(T_k) &= D_n(0) \exp\left\{\int_0^{T_k} \gamma(s) dW(s) + \sum_{j=1}^n \min\{\sigma_k, \sigma_{j-1}\} - \frac{\sigma_k}{2}\right\} \\ &\cong D_n(0) \exp\left\{\sum_{i=0}^k Z_i + \sum_{j=1}^n \min\{\sigma_k, \sigma_{j-1}\} - \frac{\sigma_k}{2}\right\}. \end{aligned}$$

In further part we will present in a boring and rigorous way what is commonly known as pricing on trees. In order to determine approximate value of the function V let us define:

1. the function $F: \mathbf{R}_+ \times \{\dots, -2a, -a, 0, a, 2a, \dots\} \rightarrow \mathbf{R}$ as

$$F(i, k) = C(T_i, \Lambda(k, i)),$$

where $\Lambda(k) = \{\Lambda_1(k, i), \dots, \Lambda_N(k, i)\}$ and

$$\Lambda_m(k, n) = D_n(0) \exp\left\{k + \sum_{j=1}^n \min\{\sigma_m, \sigma_{j-1}\} - \frac{\sigma_m}{2}\right\}$$

2. the operator H as

$$HG(k, l) = p_k(G(k, l-a) + G(k, l+a) - 2G(k, l)) + G(k, l).$$

Notice that

$$\Lambda_{m+1}(k, m) = D_{m+1}(0) \exp\left\{k + \sum_{j=1}^{m+1} \sigma_{j-1} - \frac{\sigma_m}{2}\right\}.$$

Using dynamic programming define recursively

1. $U(N_0, k) = F(N_0, k),$
2. $U(i, k) = \max \left\{ F(i, k); \frac{HU(i+1, k)}{\Lambda_{i+1}(k, i)} \right\} \text{ for } n \leq i < N_0,$
3. $U(i, k) = \frac{HU(i+1, k)}{\Lambda_{i+1}(k, i)} \text{ for } 0 \leq i < n,$

By standard optimal stopping theorems:

4. $V(0, D_1(0), \dots, D_N(0)) \cong B(0, T_0)HU(0, 0).$

5. Recombining trees for BGM model

Completely analogous procedure can be performed for single-factor BGM model, where forward LIBOR rates satisfy:

$$dL_n(t) = \left(\sum_{\substack{j=1 \\ \frac{T_j - t}{\delta} > 1}}^n \frac{\delta L_j(t)}{1 + \delta L_j(t)} \right) \eta^2(t) L_n(t) dt + \eta(t) L_n(t) dW(t).$$

An analogy of (4.1) also holds:

$$(5.1) \quad L_n(T_k) \cong L_n(0) \exp \left\{ \int_0^{T_k} \eta(s) dW(s) + \sum_{j=1}^n \min \{ \theta_k, \theta_{j-1} \} \frac{\delta L_j(0)}{1 + \delta L_j(0)} - \frac{\theta_k}{2} \right\}.$$

Construction of recombining trees (for rates and not for discount factors) and Bermudan options pricing remain unchanged. We also do not need to transform volatility as in section 2 and can directly base on market volatilities for caps and swaptions.

6. Numerical example of Polish Brady Bonds

Brady bonds are named after U.S. Treasury Secretary Nicholas Brady, who in association with the IMF and World Bank sponsored the effort to permanently restructure outstanding sovereign loans and interest arrears into liquid debt instruments. They are coupon bearing bonds with fixed, step or floating rate, having 10 to 30 year maturity. Brady bonds have semiannual interest payments, generally amortize and have the redemption option of Bermudan type. We give the example of Polish Past Due Interest (PDI) Brady Bond:

DESCRIPTION

Coupon Desc: Step-up semiannual coupon payable April 27 and October 27. 4/27/95-10/27/95: 3.25%; 4/27/96-10/27/96: 3.75%; 4/27/97-10/27/98: 4%; 4/27/99-10/27/99: 5%; 4/27/00-10/27/02: 6%; 4/27/03-maturity: 7%

Matures: 10-27-14

Country: POLAND

Date issued: 10-27-94

Guarantor: None

Issue Amount: 2,650.00 US (millions)

Currency: US

Denomination: 250,000

Amortization: Sinking fund, Semiannual payments: Oct 27, 2001 - Oct 27, 2002: 1%; Apr 27, 2003 - Apr 27, 2004: 2%; Oct 27, 2004 - Oct 27, 2009: 3%; Apr 27, 2010 - Oct 27, 2012: 5%; Apr 27, 2013 - Oct 27, 2014: 7%.

Redemption: at par at any coupon date.

Data for 20-06-2002 based on caplet volatilities

Tenor	1Y	2Y	3Y	4Y	5Y	7Y	10Y	15Y
Vols	38.80%	35.40%	30.90%	28.90%	27.00%	24.40%	22.30%	22.30%
Rates	3.35%	3.48%	3.72%	3.92%	4.08%	4.36%	4.61%	4.94%

Model parameters

Date	Nr	Coupon and sinking fund	Nominal	σ_n	θ_n
27-10-02	0	6000000	97000000	0.000%	5.31%
27-04-03	1	5395000	95000000	0.002%	12.84%
27-10-03	2	5325000	93000000	0.006%	18.59%
27-04-04	3	5255000	91000000	0.009%	23.59%
27-10-04	4	6185000	88000000	0.014%	26.33%
27-04-05	5	6080000	85000000	0.017%	28.12%
27-10-05	6	5975000	82000000	0.020%	30.32%
27-04-06	7	5870000	79000000	0.023%	32.71%
27-10-06	8	5765000	76000000	0.026%	34.48%
27-04-07	9	5660000	73000000	0.028%	36.00%
27-10-07	10	5555000	70000000	0.031%	37.37%
27-04-08	11	5450000	67000000	0.033%	38.68%
27-10-08	12	5345000	64000000	0.035%	39.98%
27-04-09	13	5240000	61000000	0.036%	41.29%
27-10-09	14	5135000	58000000	0.039%	42.62%
27-04-10	15	7030000	53000000	0.042%	43.96%
27-10-10	16	6855000	48000000	0.043%	45.31%
27-04-11	17	6680000	43000000	0.044%	46.65%
27-10-11	18	6505000	38000000	0.044%	47.99%
27-04-12	19	6330000	33000000	0.044%	49.33%
27-10-12	20	6155000	28000000	0.052%	51.48%
27-04-13	21	7980000	21000000	0.057%	53.97%
27-10-13	22	7735000	14000000	0.060%	56.46%
27-04-14	23	7490000	7000000	0.062%	58.94%
27-10-14	24	7245000	0		

Prices (without credit risk)

Model	Number of tree	Nominal	Price	Price without	Discounted
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	nodes per annum			option	cash flows
HJM	2	1 000 000.00	1 022 497.44	1 123 614.84	1 123 567.93
HJM	6	1 000 000.00	1 019 224.34	1 123 619.16	1 123 567.93
BGM	2	1 000 000.00	1 011 458.06	1 125 463.30	1 123 567.93
BGM	6	1 000 000.00	1 012 197.43	1 125 719.50	1 123 567.93
Market price	-	1 000 000.00	1 000 500.00	-	-

Results are acceptable. The program is so fast that it can be plugged into Monte Carlo for VaR. It should be noted here that BGM model slightly overprices fixed income securities – because of violation of the martingale property by the drift approximation in (5.1).

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